Detecting Problematic Argumentative Techniques: Evaluating a Frozen Configuration and a Negation Verifier on a Held-Out Dataset

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Abstract

Automatically detecting problematic argumentative techniques requires distinguishing relevant instances from expressions that resemble them lexically but do not instantiate the same argumentative mechanism. This distinction is particularly important in the presence of negation, where mentioning an equivalence relation does not necessarily amount to asserting it. This article evaluates a detection configuration, designated v2, on an author-constructed held-out dataset of 120 examples, separate from the development data. The dataset contains 60 positive examples and 60 controls across six classes. The acceptance criteria established before evaluation required no more than 22 false alarms on controls and at least 48 correct detections on positive examples. The configuration produced 16 false alarms and 54 correct detections, meeting both criteria. A negation verifier subsequently applied to the saved predictions eliminated all seven false alarms in the subset of controls involving negated comparisons, without losing any correct positive detections or changing predictions outside the equivalence class; the seven corrected predictions became correct rejections. The combined configuration therefore produced nine false alarms while maintaining 90% recall. These results establish that the frozen configuration met the predefined acceptance criteria on the evaluated dataset. They do not constitute independent validation or demonstrate broad generalization, because the dataset and reference labels were constructed by the author.

Keywords: argumentative technique detection, negation, false alarms, held-out evaluation, abstention, predefined acceptance criteria.

1. Introduction

A system that detects problematic argumentative techniques must distinguish between an actual instance of a technique and the mere occurrence of expressions associated with it. A text may mention a comparison in order to reject it, describe an argument without endorsing it, or use a question without instantiating the mechanism targeted by the detector. Evaluation must

therefore examine more than the identification of positive examples. It must also include closely related controls capable of exposing false alarms and limitations in automated decision-making.

This study addresses two clear questions. Question 1: on an evaluation set separate from the development data, does the frozen v2 configuration meet the pre-set acceptance criteria — that is, at most 22 false alarms out of 60 controls and at least 48 correct detections out of 60 positive examples? Question 2: applied on top of v2's already-saved predictions, does a negation-focused verifier reduce false alarms on negated-comparison controls, without losing correct detections on the positive examples and without changing predictions in other classes? The evaluation has a bounded objective: to test explicit criteria on examples not used to develop the evaluated configuration, rather than to demonstrate a general capacity for argumentative analysis.

2. Data and Evaluation Protocol

2.1. The six classes (operational definitions)

The classes follow the taxonomy of propaganda techniques (Da San Martino et al., 2020) and pragmatics: presupposition (a contested claim encoded as already given); manufactured equivalence (two unlike things asserted to be identical); false dilemma (a choice collapsed to two options when more exist, or unjustifiably presented as closed); whataboutism (a criticism deflected by pointing at the opponent); loaded rhetorical question (a question that asserts its own answer); implied causation (causation inferred from mere sequence or co-occurrence). A device is counted only if the author endorses it: a comparison that is negated, quoted and contested, or hedged does not count.

2.2. Held-out dataset

The evaluation dataset contains 120 examples: 60 positive examples and 60 controls, six classes each containing ten positive examples and ten controls. The author constructed the examples for a separate evaluation using topics different from those represented in development. This separation permits testing on new texts, but it does not establish methodological independence: example selection, construction, and labeling originate from the same source. The labels are therefore treated as the evaluation's operational references, not as judgments confirmed through agreement among independent annotators.

2.3. Frozen configuration

The v2 configuration was fixed before evaluation (model gpt-4o-mini, temperature 0, tag explanation-v2-frozen). Two acceptance thresholds were established: no more than 22 false alarms among the 60 controls; at least 48 correct detections among the 60 positive examples.

These are project-specific operational thresholds. Meeting them is not equivalent to passing a statistical significance test and does not imply suitability for every application.

2.4. Negation verifier

The verifier targets false alarms in cases where a text negates the equivalence relation relevant to detection. It was evaluated by applying it to saved predictions, enabling a paired comparison before and after verification on the same examples, without re-classification. The predefined criteria were: a reduction of at least 75% in false alarms within the negated-comparison subset; the loss of no more than one correct positive detection; no prediction changes outside the equivalence class. The subset contains ten controls. The baseline configuration produced seven false alarms within this subset, providing observed errors against which the intervention's effect could be measured.

2.5. Metrics and abstention

Recall is the number of correct detections divided by the 60 positive examples. The false-alarm rate is the number of incorrectly flagged controls divided by the 60 controls. Abstention must be distinguished from correct rejection: a control that does not trigger an alarm is not necessarily correctly classified, because the system may decline to decide. Consequently, the complement of the false-alarm rate is not interpreted here as the correct-rejection rate.

3. Results

3.1. Evaluation of v2

Metric	Predefined criterion	Result	Outcome
False alarms on controls	≤ 22/60	16/60	Met
Correct detections on positives	≥ 48/60	54/60	Met

These results correspond to a 26.7% false-alarm rate and 90% recall. They support the conclusion that v2 passed the operational evaluation defined for this dataset. Without additional evidence, they do not establish robustness across unexamined text populations or contexts.

3.2. Evaluation of the negation verifier

Metric	Predefined criterion	Result	Outcome
False-alarm reduction (negated-comparison subset)	≥ 75%	100%: from 7 to 0	Met
Correct positive	≤ 1	0	Met

detections lost			
Predictions changed outside equivalence	0	0	Met

The 100% reduction refers to the seven false alarms initially observed; it is not an estimate of the verifier's universal effectiveness.

3.3. Combined configuration

Metric	v2	v2 + verifier
False alarms on controls	16/60	9/60
False-alarm rate	26.7%	15.0%
Correct detections on positives	54/60	54/60
Recall	90.0%	90.0%

Across the full control set, the relative reduction in the number of false alarms was 43.8%, and the absolute decrease in the false-alarm rate was 11.7 percentage points. The individual-prediction log (the pred_final column) confirms that all seven eliminated alarms became correct rejections (the label "none"), not abstentions; the reduction was therefore not obtained by declining to decide.

4. Error and Abstention Analysis

4.1. Presupposition

The presupposition class yielded four correct detections among ten positive examples, corresponding to 40% recall within that class. This identifies a substantial limitation that should not be obscured by the aggregate recall of 90%: strong overall performance does not imply uniform performance across classes. The cause, identified from the texts, is that the presuppositions are embedded in noun phrases with loaded adjectives, which v2's stricter target reads as description rather than as a device the author endorses.

4.2. Rhetorical questions

No false alarms were reported among the ten rhetorical-question controls, but the system abstained on all ten. This behavior limits incorrect flagging, but it does not demonstrate an ability to reject controls correctly. Evaluation must preserve the distinction between avoiding an error and resolving the classification task.

4.3. Errors remaining after verification

By predicted label, the nine remaining false alarms are six false dilemmas and three implied causations. Two of the implied-causation cases come from whataboutism pairs (W04 and W08, where the “so we…” connective was read as causal), and the ninth is an implied-causation control (C06). None belongs to the equivalence class. The results show that the negation intervention does not address every source of false alarms; they motivate investigating additional verification mechanisms, but do not establish those mechanisms’ effectiveness in advance.

5. Discussion

5.1. What the evaluation supports

The primary finding is that the configuration met criteria established before the held-out results were examined. This ordering reduces the risk of retrospectively adapting the definition of success to the observed performance. The paired comparison also shows that applying the verifier produced the targeted reduction in false alarms, with no observed cost to recall and no changes outside the targeted class. The warranted conclusion is local: the intervention worked according to the predefined criteria on this dataset; it does not demonstrate that the verifier handles every form of negation or every problematic equivalence relation.

5.2. Separation from development is not independence

A dataset can be separate from development without being independent of the system’s author. Construction choices may reproduce the definitions, styles, and assumptions used during development, even when the examples and topics are new. The evidence therefore supports an author-constructed held-out evaluation, not external validation of the system.

5.3. Subsequent use of the dataset

Examining errors for reporting purposes does not invalidate the result obtained by the frozen configuration. However, if those errors are used to modify the system, the dataset becomes a development resource for the next version, and any subsequent claim about performance on data not used for adjustment would require another evaluation dataset.

6. Limitations

The evaluation has the following principal limitations. Author-constructed data and labels, without confirmation through independent human annotation. Small sample size: ten examples per condition and class do not permit robust characterization of all relevant variation. Constructed examples rather than a representative usage sample, so observed performance does not directly estimate performance on naturally occurring text streams. Artificially balanced

class distribution, whereas the prevalence of the targeted techniques may differ in actual use. Traceability requirement: reproducibility requires access to the evaluated version, protocol, labels, and individual predictions, all available in the repository. The absence of external annotators does not prevent reporting these results; it does, however, limit the interpretation of the reference labels and of performance measured against them.

7. Future Work

Three directions are warranted. First, external annotators could assess the reference labels when resources permit, clarifying ambiguous cases and measuring agreement on the operational definitions. Second, evaluation should be extended to independently collected data and naturally occurring texts while preserving the separation between development and testing. Third, the remaining errors warrant investigation, particularly in the false-dilemma and presupposition classes; any modifications informed by the present errors should subsequently be evaluated on different data, rather than reusing this dataset as untouched validation evidence.

8. Conclusion

The v2 configuration met the predefined criteria on the held-out dataset, producing 16 false alarms among 60 controls and 54 correct detections among 60 positive examples. The negation verifier also met its predefined criteria: it eliminated the seven false alarms in the negated-comparison subset without losing correct positive detections or changing predictions outside the equivalence class, and the corrections became correct rejections. The combined configuration reduced the false-alarm rate from 26.7% to 15.0% while maintaining 90% recall. These results document that the acceptance criteria were met on an author-constructed dataset separate from development. They do not demonstrate broad generalization or constitute independent validation. The contribution is a bounded, verifiable finding: a targeted intervention reduced the intended errors on new examples without observed regression in positive detection, under the conditions and limitations of the reported protocol.

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